

Swapping AttnGrounder's Attention for a Brain-Inspired Associative Memory (BDH)

Testing the “Grounding = In-Context Retrieval” Hypothesis on Talk2Car

Sudarshan Sudhakar

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Outline

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The Task: Visual Grounding

Visual grounding (Talk2Car): given an image and a natural-language command, localize the single object the command refers to.

“Turn left behind the silver car” → one bounding box.

The hard cases are scenes with **multiple objects of the same class** — the model must disambiguate using language, not just detect.

Baseline model

AttnGrounder (Mittal, IIT Mandi) — single-stage, Darknet-53 + BiLSTM/GloVe, a lightweight *visual-text attention* module fuses the two modalities before a YOLO head.

Why Bring in BDH?

Dragon Hatchling (BDH) — Kosowski et al., Pathway 2025

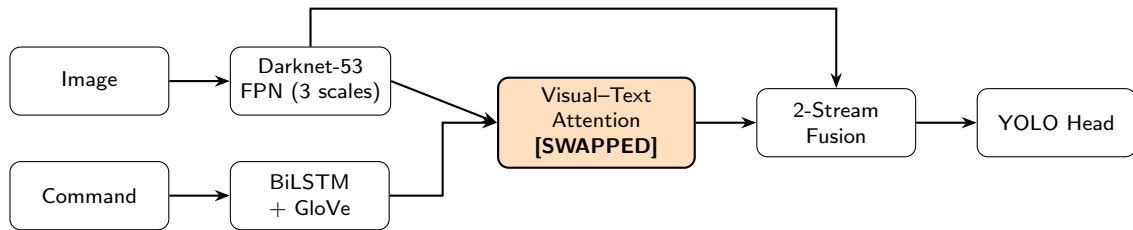
A brain-inspired sequence architecture whose working memory is an **explicit, growing outer-product associative memory** ($\rho = K^\top V$), updated with Hebbian-style read/write — rather than a fixed-size compressed state (e.g. Mamba/SSMs).

The hypothesis we're testing

“Grounding = in-context retrieval”: architectures with explicit associative memory should have an edge precisely when a query must be matched against **multiple similar candidates** in context — because that is a retrieval problem, not a compression problem.

Prediction: swapping AttnGrounder’s attention for a BDH module should raise AP50 *disproportionately* on scenes with several same-class candidate objects (“ambiguous” scenes), relative to scenes with only one.

Architecture: A Drop-In Swap



Baseline attention: softmax cross-attention (region queries $\times$ word keys), scalar gate β .

BDH module (Mode C): symmetric, non-causal kernel; ReLU-sparse addressing; explicit outer-product memory $\rho = K^\top V$; multiplicative bilinear gate. Same output shapes $\Rightarrow$ true drop-in.

Keeping the Comparison Fair

- **Single component swap:** backbone, text encoder, YOLO head, losses, hyperparameters — all byte-for-byte identical between variants.
- **Fusion trim:** BDH fuses only $[f_{visu}, f_{lang_attn}]$ (2 streams) instead of the original 3, to offset most of the module's added parameters.
- **Same training recipe for both:** A100-40GB, bf16, 100 epochs, Adam (lr 10^{-4} , backbone lr/10), batch 14, poly LR schedule.

Variant	Params	vs. baseline
Baseline (AttnGrounder, unmodified)	75.84M	—
BDH (Mode C, single-head)	76.63M	+0.79M (+1.0%)

Backbone dominates parameter count either way — this is a fusion-mechanism comparison, not a capacity comparison.

Step 6: Stratifying the Val Set

Talk2Car's own annotations only record the *referred* object — not what else is in the scene. To test the hypothesis we need to know, per image, how many **other** objects share the referred object's class.

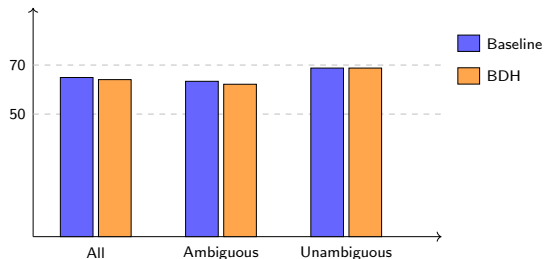
Data pipeline built for this

Join Talk2Car commands → nuScenes v1.0-trainval metadata (front-camera 3D annotations) via the sample token → count same-class objects visible in CAM_FRONT.

Bucket	Definition	Count (val, $n=1163$)
Ambiguous	≥ 1 other object, same class	830
Unambiguous	referred object is unique in class	333

Results: Aggregate & Stratified AP50

	AP50 (all) <i>n</i> =1163	AP50 (ambiguous) <i>n</i> =830	AP50 (unambiguous) <i>n</i> =333
Baseline	64.92	63.37	68.77
BDH (Mode C)	64.06	62.17	68.77
Δ (BDH – baseline)	-0.86	-1.20	0.00



Interpretation

- Both variants tie **exactly** on unambiguous scenes (229/333 correct, either model) — the entire aggregate gap sits in the ambiguous bucket.
- But the gap runs **opposite** to the hypothesis: BDH loses *more* ground precisely where its associative memory was predicted to help.

Current read

Under this configuration (Mode C, single-head, one training run per variant), the “grounding = in-context retrieval” prediction is **not supported**.

Caveats — not a final verdict yet

- Single seed per variant — a ~ 1 AP50 gap on 830 images carries meaningful noise.
- Only one of three implemented BDH kernel formulations has been trained end-to-end so far.

Next: Ablation Study

Two alternative BDH formulations are already implemented in code, but not yet trained end-to-end — this is the immediate next step.

Mode	Description
C (done)	Symmetric, non-causal — regions as queries, words as keys/values.
A (planned)	Causal-context ablation — restricts what the associative memory can attend to.
B (planned)	Single-causal-sequence ablation — words and regions share one causal stream.

Why this ordering

A null result that survives **three** kernel variants is far more defensible than one run of one variant — and both ablations reuse the exact same training/eval pipeline already built and validated.

After that: if any variant shows a real ambiguous-scene advantage, re-run with a second seed to confirm the effect is stable before drawing conclusions.

Why Keep Pursuing This

- ① **The infrastructure cost is already paid.** Data pipeline, drop-in BDH module, training loop, and the ambiguous/unambiguous stratified evaluator are all built and validated end-to-end. Remaining work is cheap experiment iteration (A100 time), not engineering.
- ② **Low marginal model cost.** BDH adds $\sim 1\%$ parameters — this stays a clean, fair test of the *mechanism*, not a bigger-model comparison.
- ③ **A well-tested negative result is still a contribution.** It would say explicit Hebbian associative memory does not trivially transfer from language modeling (where BDH was proposed) to grounded multimodal disambiguation — a useful, non-obvious data point for whether these architectures generalize.
- ④ **Directly tests a live hypothesis** from recent literature (BDH / Dragon Hatchling; the in-context-retrieval framing) in a task setting the original work did not evaluate.

Summary

- Built a validated, drop-in BDH swap for AttnGrounder and a from-scratch reproduction of the baseline (64.89 AP50 vs. paper's 63.30%).
- Built a nuScenes-based pipeline to stratify Talk2Car val by scene ambiguity (830 ambiguous / 333 unambiguous).
- First result (Mode C): hypothesis **not** supported — BDH trails baseline specifically on ambiguous scenes.
- Next: train Modes A/B on the same pipeline to see whether this is specific to Mode C's kernel or a broader finding.

Thank you — questions welcome.